

DATA 266 — Homework 6

Sneha Singh

SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: stl10_ssl.ipynb. Dataset: STL-10. Backbone: ResNet-18, no pretrained weights.

Same SID/SEED as HW1. No extra HP_ID network. The three trained models are supervised, rotation SSL, and SimCLR.

I ran this on Colab GPU (cuda). SimCLR uses the four augmentations from Demo 6.

Part A — Supervised, 10% labels

500 images from the 5,000-image train split, index list saved with seed 670. End-to-end ResNet-18, 12 epochs, Adam, lr 0.001, batch 64. Test set is all 8,000 images.

Train loss went from 2.1182 to 0.1047. Test accuracy ended at **0.3426**. That was the highest epoch in this run.

Part B — Rotation

All 100,000 unlabeled images. Predict 0°, 90°, 180°, or 270°. 15 epochs. Rotation loss went from 0.9340 to 0.2880.

I removed the 4-way head, froze the encoder, and trained a linear layer on the same 500 images for 15 epochs. Trainable weights in that step: 5,130. Test accuracy **0.4260**.

Part C — SimCLR

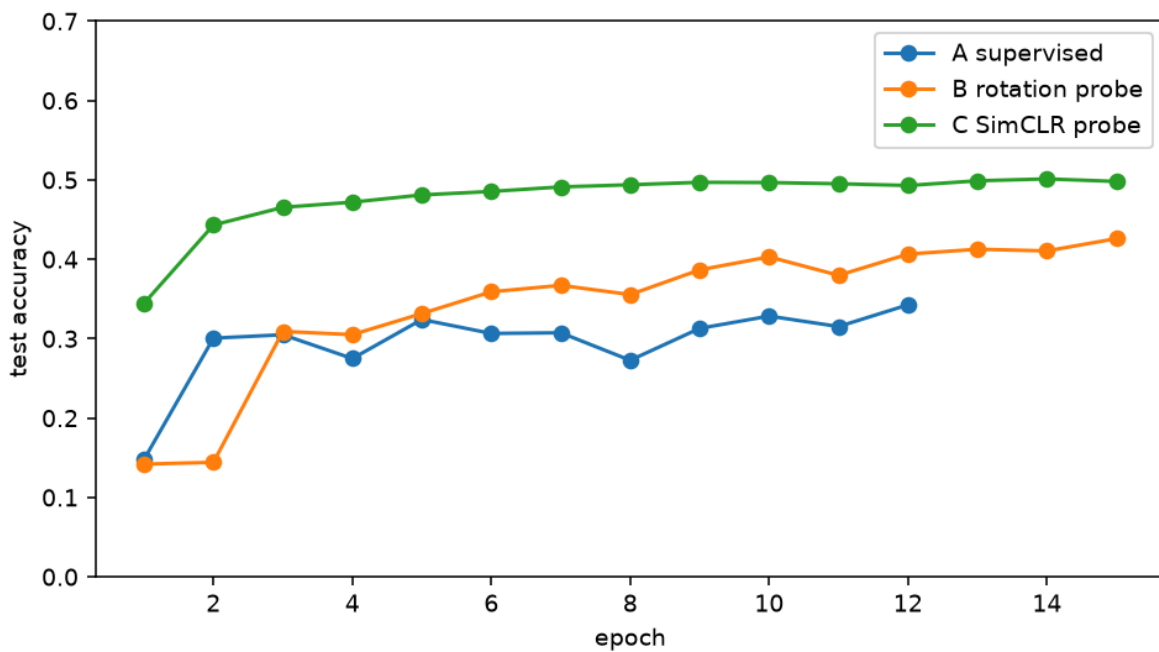
20,000 unlabeled images. Two views. Augmentations from Demo 6: random resized crop (scale 0.6 to 1.0, size 96), horizontal flip, color jitter, random grayscale. Projection MLP 512 → 128 → 128. Cosine similarity, $\tau = 0.2$. 15 epochs. Contrastive loss went from 3.3600 to 1.6038.

I removed the projection head, froze the encoder, and trained a linear layer on the same 500 images for 15 epochs. Test accuracy **0.4980**. Epoch 14 of the probe was 0.5012.

Comparison

Test accuracy is on all 8,000 test images. Parts A, B, and C use the same 500 labeled images. For B and C the encoder stays frozen and only the linear layer trains.

Model	Test accuracy
A supervised	0.3426
B rotation	0.4260
C SimCLR	0.4980



Test accuracy

Which model performed the best

SimCLR is the best. It trained on 20,000 unlabeled images, two augmented views each, cosine similarity, $\tau = 0.2$. The projection head comes off before the linear probe. Those 512-d features already separate classes better than the other two encoders, so a linear layer on 500 labels reaches 0.4980. Contrastive loss was still falling at epoch 15 (3.3600 to 1.6038).

Why the other models performed this way

Rotation is in the middle, 0.4260. It saw all 100,000 unlabeled images, and the label was only the rotation: 0° , 90° , 180° , or 270° . That beats a supervised net trained on 500 class labels. Predicting which way is up does not have to separate a dog from a monkey. Rotation loss was still falling at epoch 15 (0.9340 to 0.2880).

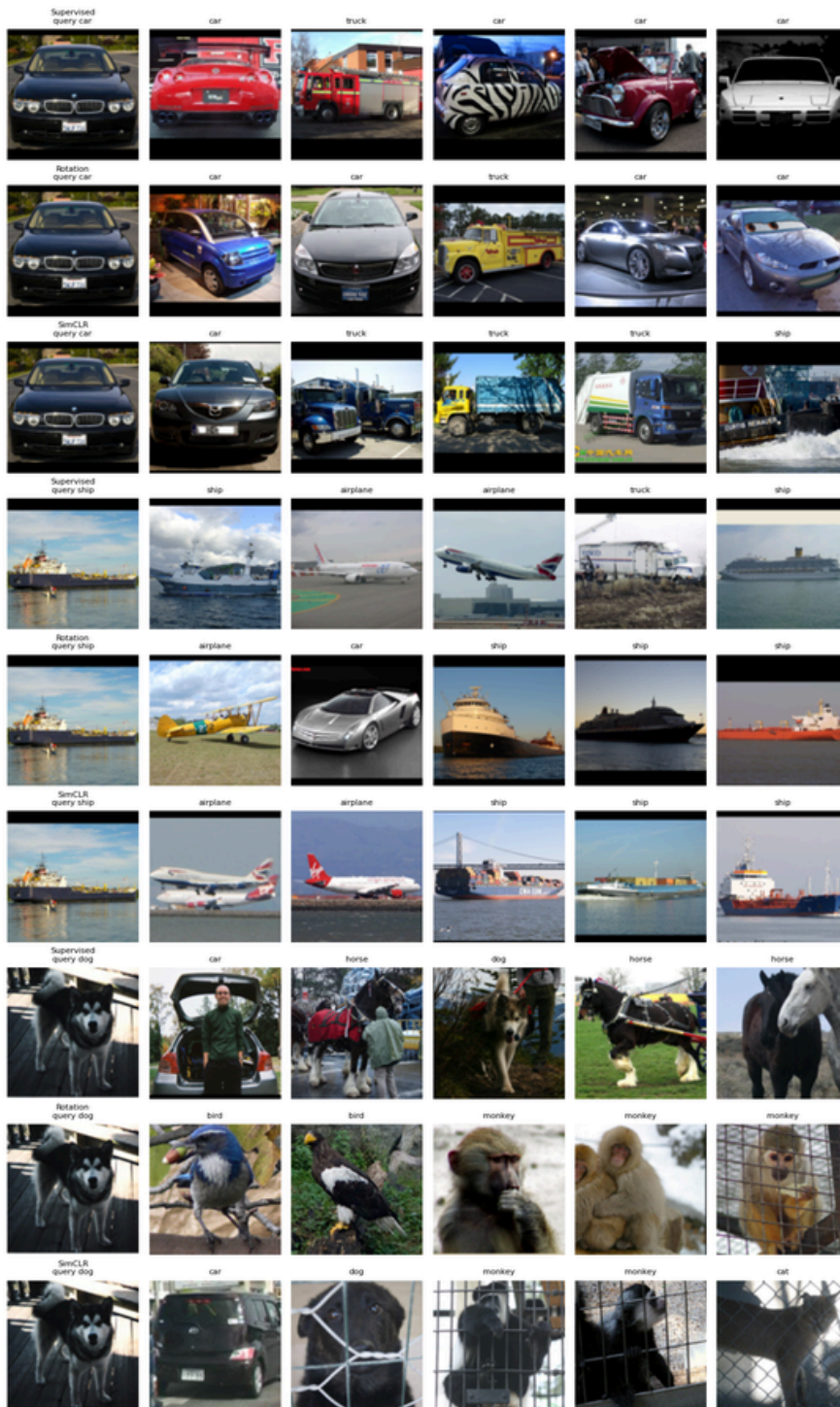
Supervised is last, 0.3426. It is the only encoder that saw class labels, and it only saw 500 images. Train loss ended at 0.1047. Test accuracy was highest on the last epoch. It fit the small labeled set.

What I would change

- Supervised: augment those 500 images. More than 10% of the labels would matter more than extra epochs. This run already peaked at epoch 12.
- Rotation: add color jitter so the encoder cannot ignore color, and train the linear probe longer. Probe loss was still 1.5839 at epoch 15.
- SimCLR: use all 100,000 unlabeled images, train past 15 epochs, and raise the batch above 128. This loss wants a big batch, and the loss was still going down.

What the nearest neighbors show

Same three test queries for every encoder: index 3806 (car), 1720 (ship), 2051 (dog). Top 5 are cosine similarity in the 512-d encoder features.



Nearest neighbors

The car query is clean for supervised and rotation: 4 of 5 neighbors are cars, and the other is a truck. SimCLR returns one car, three trucks, and a ship. Those are large vehicles, so the features are related, and the class is wrong.

The ship query flips. Rotation and SimCLR each return 3 ships. Supervised returns 2 ships, then airplanes and a truck.

The dog query is weak for all three. Supervised returns one dog, three horses, and a car. SimCLR returns one dog, two monkeys, a cat, and a car. Rotation returns two birds and three monkeys, and no dog. Close-up animal faces sit next to each other.

The probe and the grid disagree on the car. SimCLR has the highest test accuracy, and its car neighbors are the loosest of the three. I trust the 8,000-image probe for which model is best. The grid shows the features still mix classes that look alike, especially animals.